

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A company opens an account with an initial balance of **\$36,100.00**. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function ***A***, where ***A*(*t*)** is the account balance, in dollars, ***t*** years after the account is opened. The account balance after **13** years is **\$68,071.93**. Which equation could define ***A***?

Answer

- A. ***A*(*t*) = 36,100.00(1.05)^{*t*}**
- B. ***A*(*t*) = 31,971.93(1.05)^{*t*}**
- C. ***A*(*t*) = 31,971.93(0.05)^{*t*}**
- D. ***A*(*t*) = 36,100.00(0.05)^{*t*}**

Correct Answer: A

Rationale

Choice A is correct. Since it's given that the account balance, ***A*(*t*)**, in dollars, after ***t*** years can be modeled by an exponential function, it follows that function ***A*** can be written in the form ***A*(*t*) = *Nr^t***, where ***N*** is the initial value of the function and ***r*** is a constant related to the growth of the function. It's given that the initial balance of the account is **\$36,100.00**, so it follows that the initial value of the function, or ***N***, must be **36,100.00**. Substituting **36,100.00** for ***N*** in the equation ***A*(*t*) = *Nr^t*** yields ***A*(*t*) = 36,100.00*r^t***. It's given that the account balance after **13** years, or when ***t* = 13**, is **\$68,071.93**. It follows that ***A*(13) = 68,071.93**, or **36,100.00*r¹³* = 68,071.93**. Dividing each side of the equation **36,100.00*r¹³* = 68,071.93** by **36,100.00** yields ***r¹³* = $\frac{68,071.93}{36,100.00}$** . Taking the **13**th root of both sides of this equation yields ***r* = $\sqrt[13]{\frac{68,071.93}{36,100.00}}$** , or ***r*** is approximately equal to **1.05**. Substituting **1.05** for ***r*** in the equation ***A*(*t*) = 36,100.00*r^t*** yields ***A*(*t*) = 36,100.00(1.05)^{*t*}**, so the equation ***A*(*t*) = 36,100.00(1.05)^{*t*}** could define ***A***.

Choice B is incorrect. Substituting **0** for ***t*** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**.

Choice C is incorrect. Substituting **0** for ***t*** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**. Additionally, this function indicates the account balance is decreasing, rather than increasing, over time.

Choice D is incorrect. This function indicates the account balance is decreasing, rather than increasing, over time.